

Conductor-Functorial Borel Torsors

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Summary

Pass 90 makes the Pass-89 Borel-torsor theorem functorial across conductor and radical supports. The key correction is directional: support restriction is canonical, but support enlargement is not a canonical map on the diagonal quotient torsor.

For a finite prime support S , write

$$P(S) = \left(\prod_{p \in S} \mathbb{Z}_p \right) / \Delta \mathbb{Z}.$$

If $S \subseteq T$, coordinate projection descends to a well-defined restriction $P(T) \rightarrow P(S)$. Zero-insertion $P(S) \rightarrow P(T)$ does not descend when new primes are added, because a diagonal shift in S maps to a vector with zeros in the new coordinates rather than to a diagonal shift in T .

Main Claims

- **Support restriction.** For $S \subseteq T$, projection gives a canonical map $P(T) \rightarrow P(S)$.
- **No canonical insertion.** If $T \setminus S$ is nonempty, zero-insertion is a representative-level choice, not a quotient-level homomorphism $P(S) \rightarrow P(T)$.
- **Meet/join comparison.** Rad-incomparable supports compare by the meet span $P(S) \rightarrow P(S \cap T) \leftarrow P(T)$ and by a join arena $P(S \cup T)$ for gluing.
- **Finite Borel naturality.** The finite shadows $(\mathbb{Z}/N)^\times \ltimes \mathbb{Z}/N$ reduce along $N \mid N'$, preserving the unit class and singleton strict marked stabilizer.

Machine Verification

`code/scripts/check-pass90.py` produced `artifacts/reports/pass90-conductor-functorial-borel-torsors-check.json` with overall PASS. It checks radical support invariance, projection descent, failure of zero-insertion under new primes, meet/join comparison rows, and finite Borel reductions along conductor divisibility.

Next Use

Pass 91 should decide whether the restriction/span Borel-torsor package is a sheaf, stack, prestack, or descent-obstruction object over the finite prime-cover site.